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CHAPTER 14

Vocal Communication of Emotion



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FROM EXPRESSION TO COMMUNICATION

There is a growing consensus among emotion psychologists that emotion needs to be viewed as a multicomponent entity (Frijda, 1986; Lang & Cuthbert, 1984; Lazarus, 1991; Lewis & Michalson, 1983; Scherer, 1984; Smith & Ellsworth, 1985). Rather than treating subjective experience or feeling as a synonym of emotion, psychologists now generally consider it just one of several components of the total emotion construct. Two other major components are neurophysiological response patterns (in the central and autonomic nervous systems) and motor expression (in face, voice, and gesture). These three components—feeling, physiology, and expression—are often referred to as the “emotional reaction triad.” Many theorists favoring a componential view include the evaluation or appraisal of the antecedent event and the action tendencies generated by the emotion, as components of the emotional process (see Frijda, 1986; Lazarus, 1991; Scherer, 1984; Smith & Ellsworth, 1985).

In considering emotion as a phylogenetically evolved, adaptive mechanism that facilitates an organism’s attempt to cope with important events affecting its well-being, Scherer (1984, 1993) has proposed that changes in the respective components are integrated or synchronized within the emotion episode. This proposal is

based on the assumption that all the resources of an organism need to be mobilized, and all systems coupled, in order to maximize the organism’s chances to cope with an important event. Since all components interact with each other, changes in one component will produce changes in another. For example, the elevated respiration rate required for increased oxygen supply under arousal will affect facial expression (mouth shape) and vocal expression (changes in subglottal pressure) as well as a number of peripheral physiological parameters.

Ever since Darwin’s (1872) pioneering work on the expression of emotions in humans and animals, expressive behaviors have been explained as consequences of other adaptational responses (as described above) or as functional responses in their own right (e.g., facial muscle movements to block odor intake or to sharpen the vision; preparatory movements for intended action; see Ekman, 1972; Ekman & Friesen, 1975; Frijda & Tcherkassof, 1997; Scherer, 1992; Smith & Scott, 1997). However, the motor expression component occupies a special role in this adaptational context: It extends the adaptational response beyond the boundary of the individual and his or her action potential, by communicating the individual’s reaction to an event and intention to act in a certain way (the action tendency) to the social environment. Once expressive motor behavior starts to serve as a signal, there will be selective shaping of

functionally based movements for the purpose of communication, through processes such as ritualization, formalization, or symbolization (see Chevalier-Skolnikoff, 1973; Eibl-Eibesfeldt, 1984; Leyhausen, 1967; Redican, 1982; Schneider & Dittrich, 1990; Van Hooff, 1972).

Scherer and his collaborators have suggested acknowledging the dual determinants of expressive signals by differentiating between "push effects," in which physiological processes such as respiration and muscle tone push expressions in a certain direction, and "pull effects," in which external factors such as social norms or listener expectations pull expressions in a certain direction (Scherer, Helfrich, & Scherer, 1980). In the case of push effects, internal support and action subsystems will affect the production of vocalization in an essentially nondirective manner. For example, increased muscle tension produced by ergotropic arousal can affect breathing patterns, the functioning of the vocal folds, the shape of the vocal tract, and facial expression. In addition, the behaviors resulting from a particular emotional state, such as threat postures and rapid movement, will affect vocalization (Zahavi, 1982). Thus, as muscle tone is likely to be higher in sympathetic arousal, the fundamental frequency of the voice (F_0) will also be higher. Pull effects, on the other hand, are governed by physical communication conditions (e.g., noise), social norms or conventions such as display rules, listener expectations, or other context factors. These pull effects influence the production of signs by specifying a particular motor target pattern, as opposed to mental concepts or internal physiological processes that push out expression (for detailed discussions of push and pull effects, see Scherer, 1985; Scherer & Wallbott, 1990). Of course, once a signal has been symbolized through consistent shaping by pull effects, it acquires representational status.

To describe the functions of emotional expression, Scherer (1992) has suggested using the "organon" model of Bühler (1934). In this model, an expressive behavior has three functions: as a symbol representing the component of emotion it stands for; as a symptom of the state of the expressor; and as an appeal, or signal, designed to elicit a response from the receiver(s). The vervet monkey alarm call (Marler, 1984), for example, serves all three of these functions: It functions as a symbol of different

predators, as a symptom of the fear state of the animal, and as an appeal to others to run away. Furthermore, these functions are mutually interdependent. For example, if a call refers to an air predator, both the emotional reaction and the appeal may be very different from those made in reference to a ground predator. In the first case, both emitter and receiver might seek shelter under a bush and freeze; in the second, they might become highly activated and run up a tree.

The tripartite nature of emotional expression—serving as a symptom of changes in organismic subsystems; as a communicative act; and, in its symbolized form, as a representational sign—is at the root of many of the current debates about the nature of expression. Much confusion can result from overemphasis on one of these three functions. For example, Dawkins and Krebs (1978) and Caryl (1979) have highlighted the appeal function, claiming that most if not all expressive behavior can be explained as strategic intention signaling. More recently, Fridlund (1994), opposing a "readout view" of expression, has reiterated the concept of strategic intention movements, in what he calls a new "behavioral ecology view" of expression. Such views appear to neglect the involuntary physiological changes accompanying emotional arousal—changes that affect facial and vocal expression independently of any intentional effort (e.g., the respiration changes due to stage anxiety will affect a speaker's glottal pulses and thus change voice quality, in spite of powerful intentions to keep the voice straight). Furthermore, the claim that virtually all expressive behavior is exclusively determined by strategic intentions and does not provide any reliable information about underlying states has been strongly challenged by Hinde (1981), who has shown that there are many situations in which truthful signaling is clearly advantageous to the sender. In other cases, where there is a conflict between several behavioral alternatives (e.g., fight and flight), maximum ambiguity in the meaning of the expressive display, rather than outright deceit, would seem to be the optimal strategy. Similarly, Zahavi (1982) has shown that an individual who feels superior in an agonistic encounter does not gain by deception, whereas the weaker individual actually stands to lose: Producing the deceptive signal is costly and may reduce the chances for successful flight (e.g., by depleting energy).

Zahavi concludes that the cost of the signal is what selects for its reliability.

One can argue that the sincerity attributed to a sender is directly correlated with the extremeness (and thus the cost in terms of both muscular exertion and social image) of an affect display, and that the closer a signal comes to being determined by push effects, the more trustworthy it is seen as being. Scherer (1994) has predicted that "raw" affect vocalizations will be interpreted as more spontaneous and reliable signals, as well as more truly felt, than will conventionalized, ritualized "vocal emblems," produced in a socially stereotyped way with "typical intensity" (cf. Hinde, 1981). For example, the pain experienced by a sender emitting an unarticulated roar seems more real to us than the one indicated by the use of conventionalized emblems such as "Ouch!" For the sake of credibility of the signaled state, pull effects cannot move too far away from the original push effects on which they are based (except, of course, for expressions with iconic or arbitrary coding of external referents).

Quite apart from cost aspects, the efficacy of strategic deceit clearly depends on the general reliability of the signal as an index of the emotional state that is actually present. For example, counterfeit money only works if the real money actually buys something. It is difficult to envisage an economy with only counterfeit money circulating. Similarly, if an expressive signal does not represent anything but the (potentially deceptive) intent of the signaler (and if, in addition, it is a cheap signal), it will quickly lose its value. Of course, the matter is frequently complicated by the fact that the emotional process underlying expressive signaling is often highly complex and even conflictual, with different action tendencies and behavioral intentions being activated, and pull effects (including those generated by strategic considerations) constantly changing the signal.

MULTIPLE DETERMINANTS OF THE NATURE OF AFFECT VOCALIZATION

In discussing the origins and functions served by emotional expression, we have touched in the preceding section upon how a range of push and pull factors shape the resulting speech signal. Things are further complicated

by the way in which the components of the human speech system subserve many other functions that are indirectly, or not at all, linked to emotional communication. How these multiple systems manage to fulfill their functions simultaneously is a largely unaddressed question. For example, in the case of respiration, it is unclear whether the respiratory requirements of speech have priority over, or are subservient to, the demands of the emotional response system. Perhaps there is a tradeoff between the two systems, whereby an effort is made to satisfy all needs, albeit suboptimally. In addressing such questions, we need to consider the multiple determinants, both push and pull, of vocal affect expression.

Physiology

The most fundamental determinants of vocal affect are the physiological changes that accompany emotions, which in turn produce changes in the functioning of the vocal production systems (see Scherer, 1986, for a thorough treatment). In summary, it is accepted by many emotion psychologists that emotions are accompanied by various adaptive responses in the autonomic and somatic nervous systems (Cacioppo, Berntson, Larsen, Poehlmann, & Ito, Chapter 11, this volume; Levenson, Ekman, Heider, & Friesen, 1992; Öhman, 1994; Smith, 1989; Stemmler, 1996). These responses will lead to changes in the functioning of parts of the speech production system, such as respiration, vocal fold vibration, and articulation. For example, with a highly aroused emotion such as rage, increased tension in the laryngeal musculature coupled with raised subglottal pressure will provoke a change in the production of sound at the glottis, and hence a change in voice quality. Whether or not the physiological responses (and hence the resulting vocal changes) that accompany emotions are emotion-specific is an ongoing topic of debate. Scherer (1986) has suggested, however, that it is feasible to formulate a number of concrete predictions about the physiological, and hence the vocal, reactions in response to appraisal profiles that generate specific needs for organismic adaptation.

Cognitive and Attentional Constraints

Certain emotional situations impose high demands on attention and cognition, with the re-

sult that speech planning and execution can be expected to suffer, leading to changes in speech fluency and prosody. The presence of new and unexpected information, leading to an orienting response and refocusing of attention (as in surprise), can cause speech errors and interruptions. Emotions such as anxiety, which entail sustained cognitive workload, adversely affect speech planning and execution (Tolkmitt & Scherer, 1986). In contrast, other emotions (e.g., contentment) may improve speech fluency through the minimizing of extraneous, distracting thoughts. In summary, the effect of any emotion on speech will depend on the attentional and cognitive conflict between the speaker's emotional response and the focus of speech. There is also evidence that focusing of attention and cognitive load are accompanied by physiological changes, such as different breathing styles (Boiten, 1993), which are likely to affect the voice.

Vocal Transmission Characteristics

One feature of vocal expression is that it is optimally suited to relatively indirect communication over long distances. Visual contact between speaker and listener is not necessary. Moreover, sound can be used to attract attention, as listeners almost always have their listening "turned on" and open to sounds coming from virtually any source (Scherer, 1994). In contrast, over long distances the voice is not a particularly private medium. One might predict that these characteristics of the vocal channel make it more suitable for the signaling of certain emotions than of others. This hypothesis is consistent with the fact that different emotions are expressed and perceived more or less easily in the voice. For example, fear and alarm are among the most clearly vocally expressed emotions. There is obvious adaptive benefit to being able to warn cohorts of danger by communicating such affective states over long distances. Disgust, on the other hand, has been found to be universally badly communicated in the voice, possibly because it is an emotion that, from an evolutionary perspective, needs to be communicated only over relatively short distances and can thus be expressed very well visually. (There are, however, other ways of expressing disgust vocally, which are intimately connected to the physiological reaction to disgusting stimuli. See Scherer, 1994, and the section on decoding studies, below.)

Role of Language

One of the most obvious, but possibly most complicated, examples of how a functional system quite distinct from emotion can nevertheless have an effect on emotional expression is the use of the voice as a carrier signal for language. Human speech has been grafted onto a phylogenetically older vocal call system that was mainly used for affective and social signaling. It is unclear to what extent the evolution of spoken language might have been constrained by the affective signaling system, or alternatively how spoken language in its present form constrains emotional expression. There is overwhelming evidence that the two systems can be dissociated; that is, they can function independently of one another (Gorelick & Ross, 1987; Lieberman & Michaels, 1962; Ross, 1984; Scherer, Ladd, & Silverman, 1984; Starkweather, 1956). Some research has shown less use of short term changes in fundamental frequency to express emotion in tone languages (in which short-term F_0 contours are used to carry lexical information) than in Indo-European languages (in which F_0 plays no lexical role) (Ross, Edmondson, & Seibert, 1986). Thus it seems that in some cases at least, use of a particular acoustic feature in spoken language limits its use for the communication of emotion.

It is also likely that, in parallel with the development of spoken language, more formal, prototypical ways of expressing emotions or at least emotional attitudes (e.g., contempt, sarcasm, and curiosity) have been established. These affective representations, which could be described as a type of "affective pragmatics," are partly determined by cultural norms, such as rules governing politeness and etiquette in speech; they thus probably vary across cultures and languages. They are also likely to be enlisted for strategic social aims, as discussed earlier in this chapter and in Scherer (1988). Such emotion prototypes however, have been based upon, and thus still reflect, original push factors (Scherer, 1986; Scherer, Banse, & Wallbott, 1999).

RESEARCH EVIDENCE ON THE ENCODING OF EMOTION

Until recently there was little systematic knowledge about the details of the encoding of emotion in the voice, in terms of either the external-

ization of underlying psychophysiological states or the realization of affective pragmatics. A growing interest in emotion psychology, coupled with progress in speech science, has led to an increasing number of studies addressing these issues. This section provides an overview of the existing literature in this area and highlights the questions that remain to be answered, as well as a number of new questions that have been brought to light.

Methodology

There is a great amount of variation in the methods that have been used to study the encoding of vocal affect (see Scherer, 1982, 1986; Wallbott & Scherer, 1986). A lack of consistency in number and type of speakers, in number and type of emotions studied, in type of stimulus materials or speech samples used, and in the nature of the context and the eliciting conditions limits the extent to which prior studies can be compared and results accumulated. In the case of choosing suitable speech materials, this reflects a lack of sophistication in designing speech experiments—a problem that can be resolved by adopting some of the standard speech material used in mainstream speech science. Speakers have ranged from university students to radio professionals, pilots, and actors, with actors being the most common. Since acted portrayals probably represent culturally shaped prototypes of affective speech, their use in encoding studies is valid; however, this needs to be supplemented by other types of affective speech occurring as a result of genuine emotions, either induced in the laboratory or observed in natural settings.

Martin (1990) provides an overview of the techniques that have been used in the induction of emotions, which include emotional stories, pictures, music, or films and mental imagery. Computer games and tasks are other possible techniques for the induction of real emotions in the laboratory and have begun to be used in various forms by researchers (Anderson & Ford, 1986; Banse, Etter, Van Reekum, & Scherer, 1996; Kaiser, Wehrle, & Edwards, 1994; Kappas, 1997; MacDowell & Mandler, 1989; Van Reekum, Johnstone, & Scherer, 1997). Computer games can be programmed to manipulate emotion-eliciting situations while controlling for extraneous factors, and are more interactive than other induction techniques. The problem with all of these methods, however, is

that ethical constraints prevent anything but the induction of low-magnitude emotions, which may not provide sufficient power for their differentiation on the basis of vocal cues. The study of "real-life" emotional speech in natural situations is difficult because of the poor acoustic recording conditions usually present. One obvious candidate for the study of fairly intense emotions is sporting events, where competitors, spectators, and commentators alike often experience a range of emotions of quite high intensity. "Real-life" and confrontational television programs are also intriguing possibilities for gathering emotional speech, although the ethics of such programs are perhaps more dubious than many laboratory studies that have been disallowed by university ethics committees.

Acoustic Measures of Emotional Speech

In principle, the encoding of emotion in the voice can be measured at any of a number of stages, from the physiological changes in various parts of the vocal production system to the perceptual judgment of vocal quality by a listener. The decision as to which stage (or stages) is most suitable depends on the specific research project and its goals. In order for researchers to be able to integrate data across both encoding and decoding studies, however, measurement of the physical (i.e., acoustic) properties of the propagating speech sound is most appropriate. Many acoustic measures of human speech and vocal sounds have fairly well-understood relationships with perceived aspects of speech. Although the relationships between the physiology of speech production and the resulting acoustic signal are less clear, there exists a large and rapidly expanding literature on the subject (see Borden & Harris, 1994, for an overview of speech production and perception). Acoustic analysis thus allows a comparison of studies of emotion encoding in speech with studies focused on the decoding or perception of expressed emotion (which will be discussed in the next section).

For the sake of our discussion here, we subdivide acoustic measures of speech into four categories: time-related measures, intensity-related measures, measures related to fundamental frequency, and more complicated time-frequency-energy measures. The first three categories are linked mainly to the per-

ceptual dimensions of speech rate, loudness, and pitch, respectively, whereas the fourth category has more to do with perceived timbre and voice quality. Since speech is a rapidly changing, dynamic signal, most standard acoustic variables are measured over very short speech segments, over which it can be assumed that the signal is relatively stable. To obtain long-term or suprasegmental measures, the short-term acoustic variables can be aggregated over longer time frames. Since emotion is assumed to be a fairly long-lasting phenomenon, most research to date has used this approach, typically aggregating variables over single or multiple phrases or sentences. As research leads to more precise hypotheses, especially concerning affective pragmatics, it will be necessary to examine emotional speech at shorter time scales. The following discussion is a brief introduction to the four acoustic categories and their associations with speech production and perception; many texts offer a more detailed treatment of the acoustics of speech (e.g., Borden & Harris, 1994; Deller, Proakis, & Hansen, 1993).

Time-Related Measures

The speech signal consists of a temporal sequence of different types of sounds (those corresponding to vowels, consonants, and interjections) and silence, all of which can be carriers of affective information. The most conceptually simple time-related measures of speech are the rate and duration of both vocal sounds and pauses, which are expected to vary for different expressed emotions. In practice, however, there exist no clear guidelines as to which basic units of vocal sound are most appropriate for the analysis of emotional speech. Past research has been limited mostly to measures of overall phrase duration and word rate. A similar problem exists for measurements of silence. It seems necessary to distinguish between different types of silence, such as micropauses that might occur between words or phrases, and longer pauses reflecting hesitations.

Intensity-Related Measures

Although the amplitude of a speech signal is a direct measure of the sound pressure, an acoustic measure that more closely reflects both the production and perception of speech is intensity. Intensity is a measure of the amount of energy in a speech signal, and therefore re-

flects the effort required to produce the speech. It is also the case that the perceived loudness of speech depends on the intensity rather than the amplitude. The standard unit used to quantify intensity is a logarithmic transform of the intensity, called the decibel (dB).

Although it is quite easy to measure, the intensity of speech is dependent on several factors other than those that are (usually) of interest. The distance and orientation of the speaker relative to the microphone, the acoustic characteristics of the recording location, and the calibration of the recording equipment must all be carefully controlled if intensity measurements are to be free of artifacts. Despite these issues, both intensity and variability of intensity have been shown to be useful acoustic variables in the measurement of affect.

Measures Related to Fundamental Frequency

One of the most frequently used vocal cues is fundamental frequency, measured in cycles per second, or hertz (Hz). F_0 is the rate at which the vocal folds open and close across the glottis, and it strongly determines the perceived pitch of the voice. It varies continuously and often quite rapidly, as a function of both linguistic and paralinguistic aspects of speech. Over the course of a phrase or utterance, F_0 can be quantified in terms of F_0 floor (the base level of F_0 during the utterance) and F_0 range (the extent to which F_0 rises above the floor level during the utterance). Other measures derived from F_0 include F_0 variance, F_0 skewness, and jitter, which is the extent to which the time for the vocal folds to open and close fluctuates rapidly and randomly from one vocal cycle to the next. Although mean F_0 has been used in a number of studies of vocal affect, its continued use is discouraged except for comparison with previous research, as its value really reflects the combined effects of F_0 floor and F_0 range.

More complicated F_0 -related acoustic variables have been proposed to examine the relative power of the fundamental frequency and the harmonics (F_1 , F_2 , etc.), which is determined by the manner in which the vocal folds open and close. Some of these, such as the ratio of F_1 to F_0 and the spectral slope of the glottal spectrum, have been used as indicators of voice quality (Klatt & Klatt, 1990). Unfortunately, measurement of these parameters necessitates

that the frequency characteristics of the vocal fold vibration be separated from those due to the vocal tract resonance; this involves an estimation process (usually inverse filtering) that is at best highly mathematically complicated, and at worst very unreliable.

Combined Time-Frequency-Energy Measures

During speech the morphology of the vocal tract is varied by placement of the various articulators (i.e., tongue, lips, teeth) in different configurations. Each articulatory configuration produces a specific pattern of resonant frequencies, thus forming the individual sounds (phonemes) of a language. It is customary to characterize each phoneme by the amplified frequencies corresponding to that phoneme, which are called "formants." Although the formants primarily carry phonetic information, they are still affected by changes to the vocal tract that accompany different emotions, such as changes to articulator muscle tension and salivation. The amount of resonance (formant amplitude) and the range of frequencies amplified in a given formant (formant bandwidth) may change considerably with different emotional states. Unfortunately, analysis of the formants of speech is a time-consuming process riddled with methodological pitfalls, and thus it has rarely been applied to the study of affective speech.

A related measure is the "long-term average spectrum" (LTAS), which is the distribution of energy over a range of frequencies, averaged over an extended time period (at least 30 seconds). Calculated separately for the voiced and unvoiced sections of speech, the LTAS is an attempt to arrive at an estimate of the long-term resonance characteristics of the vocal tract by "averaging out" all the phoneme-dependent variation, such as that measured in formant analysis. The remaining measure should reflect both speaker-dependent characteristics and resonant characteristics attributable to fairly long-term changes in speaker state, such as emotions (Pittam, 1987; Pittam, Gallois, & Callan, 1990).

Glottal waveform analysis is a technique yielding a number of parameters that characterize the sound produced at the glottis, prior to the effects of vocal tract resonance. The aim is to quantify aspects of the glottal waveform that can be directly related to the physiology and

mechanics of the opening and closing of the vocal folds (Fant, 1993). Its close relationship to glottal physiology makes it a promising candidate for studies on affective speech.

In addition to using the various types of acoustic measures individually, researchers can obtain more complex variables by combining two of the three dimensions. For example, Scherer and Oshinsky (1977) used variations in the vocal envelope (a measure that combines the amplitude and time dimensions) in studying the attribution of emotional state. Although studies are relatively scarce, there is evidence that the temporal pattern of F_0 change—that is, the form of the intonation contour, may also play an important role in the communication of affect (Cosmides, 1983; Frick, 1985; Mozziconacci, 1995, 1998; Uldall, 1960).

Evidence on Acoustic Correlates of Emotion

Space limitations prevent us from referring in detail to the previous research on the encoding of vocal affect. For more detailed reviews, including extensive references to previous studies, the reader should consult Frick (1985), Scherer (1986), and Banse and Scherer (1996). Below we provide a brief summary of the major findings for the most commonly studied emotions.

1. *Stress*. Although stress is not a single, well-defined emotion, it is useful to start this overview with a look at research on the acoustic indicators of psychological and physiological stress, which were until recently the focus of more research in speech science than specific emotions were. Conditions of high stress or high mental workload have generally been found to lead to raised values of F_0 , greater intensity, and lower utterance duration (i.e., faster speech). Stress-induced variation in the position or precision of formants has also been reported in some studies. These results are difficult to interpret, however, because changes to these parameters seem to depend on factors such as the speaker's susceptibility to and strategies for coping with stress, as well as the type of stress (i.e., cognitive or emotional; see Tolkmitt & Scherer, 1986).

2. *Anger*. In general, an increase in mean F_0 and in mean intensity has been found in angry speech. Some studies also show increases in F_0 variability and in the range of F_0 across the ut-

terances encoded. Considering that mean F_0 is not a single acoustic measure, it is not clear whether angry speech has a higher F_0 level or a wider range of F_0 or both. It is possible that those studies finding increased F_0 range and variability have measured "hot" anger or rage, whereas those studies not finding these characteristics may have measured "cold" anger or irritation, as found by Banse and Scherer (1996). F_0 contours tend to be directed downward and articulation rate tends to be generally increased for hot anger. Anger also seems to be characterized by an increase in high-frequency energy, which, together with the increase in intensity, probably reflects greater subglottal pressure leading to more energy in the higher harmonics.

3. *Fear*. Expected high arousal levels for fear are consistent with convergent evidence showing increases in intensity, mean F_0 , F_0 floor, and F_0 range, although Banse and Scherer (1996) found a decrease in F_0 range. As with anger, the increased intensity of fearful speech accompanies an increase in high-frequency energy. Rate of articulation is also higher. Related emotions such as anxiety or worry also show faster articulation, but data on the other variables are less consistent. Some studies have found an increase in mean F_0 , but Banse and Scherer (1996) reported a decrease in mean F_0 , F_0 floor, and F_0 range. A decrease in intensity for anxious speech has also been reported.

4. *Sadness*. As with fear, the findings converge across the studies that have included sadness. Decreases in mean F_0 , F_0 floor, F_0 range, and intensity are usually found (although Banse & Scherer, 1996, reported increased F_0 range), as are downward-directed F_0 contours. Corresponding to the decrease in intensity, voiced high frequency energy seems attenuated, indicating weaker higher harmonics. Rate of articulation also decreases for sadness. Most studies reported in the literature seem to have studied the quieter, more resigned forms of this emotion, rather than the more highly aroused forms such as desperation (where correlates reflecting arousal are found, such as increased intensity, increased F_0 floor, and increased high-frequency energy).

5. *Joy*. This is one of the few positive emotions frequently studied, most often in the form of elation rather than the more subdued forms, such as enjoyment or happiness. Consistent with the high arousal level that one might expect, there is a strong convergence of findings

of increases in mean F_0 , F_0 floor, and intensity. F_0 range and F_0 variability are also found to increase. There is also inconclusive evidence for an increase in high-frequency energy and an increase in the rate of articulation. Quieter forms of the emotion, such as contentment, seem to be characterised by relaxed vocal settings, leading to low mean F_0 , low F_0 floor, lower intensity, slower articulation, and weaker higher harmonics.

6. *Disgust*. As noted by Scherer (1989), the results for disgust tend not to be consistent across the encoding studies. The few studies that have included this emotion vary in their induction procedures from having an actor simulate emotion to having subjects view unpleasant films and measuring their disgust (or possibly displeasure). The studies using the former procedure found a decrease in mean F_0 , whereas those using the latter found an increase in mean F_0 . Even within studies, little consistency in the acoustic characteristics of disgusted speech has been found, implying that disgust is simply not well encoded by speakers. This conclusion is echoed in the decoding literature (see below), where disgust is universally reported to be poorly recognized in speech.

7. *Boredom*. Bored speech is generally slow and monotonous, with low F_0 floor, low F_0 range and variability, and slow rate of articulation. Interested speech tends to be the opposite, with large F_0 range and increased speaking rate.

It is evident from this brief review that where there is considerable consistency in the findings, it is usually related to arousal, regardless of the specific quality of the emotion under investigation. This should not, however, be taken as evidence that discrete emotions are not differentiated by vocal cues. Indeed (as we shall see in the ensuing section on decoding studies), given the high recognition of emotions in speech, there must exist emotion-specific acoustic patterns. There are several reasons why acoustic patterns that differentiate emotions with similar arousal have not yet been consistently found.

For one thing, very few acoustic parameters have been measured in most previous studies. Furthermore, the relatively simple acoustic measures employed preclude a finer level of emotion differentiation, because they tend to be exactly those measures that are most affected by physiological arousal. Many more precise

acoustic measures, some of which have been mentioned above in the category of "combined time-frequency-energy parameters," could be applied to the analysis of emotional speech. Indeed, ongoing research that makes use of formant analysis, spectral and cepstral analysis, intonation scoring, and inverse filtering of the speech signal to arrive at an estimate of the glottal waveform looks very promising (Bachorowski & Owren, 1995, 1996; Banse & Scherer, 1996; Klasmeyer, 1997).

A second factor limiting the identification of emotion-specific acoustic patterns has been the lack of precision in defining emotional states. As can be seen from the preceding overview, often two or more forms of the same basic, or modal, emotion exist. Examples are anger (rage vs. irritation), happiness (contentment vs. elation), sadness (disappointment vs. distress) and fear (anxiety vs. terror). Although in some sense semantically similar, the different forms often have quite different vocal profiles, neither of which will be discovered if both emotions are lumped together under one label.

The recent study by Banse and Scherer (1996) has shown that a greater range of acoustic parameters, coupled with a better specification of expressed emotions, can result in a sizeable increase in the discriminative power of the acoustic parameters. Further improvements should be possible through greater refinement of the acoustic parameters. For example, it remains unclear how the measures of average voiced spectrum used by Banse and Scherer relate to either vocal tract or glottal waveform characteristics. Measurement of formant frequencies, amplitudes, and bandwidths, as well as measurement of the glottal waveform, would provide better understanding of the underlying processes involved in emotion encoding and might also increase emotion discrimination.

A final word is warranted concerning the effects of individual and cultural differences on affective speech. Although there is overwhelming evidence from decoding studies (see below) that affective speech is quite similar across individuals and cultures, relatively little has been done to quantify these effects in vocal affect encoding. Most commonly the effects of the individual speaker have been factored out of the analyses as an unwanted source of variance. Very few cross-cultural or cross-language studies have been carried out on emotion encoding in the voice.

RESEARCH EVIDENCE ON THE DECODING OF PORTRAYED EMOTION

Accuracy of Decoding Discrete Emotions from Voice Samples

A large number of empirical studies have demonstrated that human listeners as judges can correctly recognize or infer speaker affect state or attitude from voice samples, regardless of the verbal content spoken. In the most common paradigm, speakers (often actors, either amateur or professional) are asked to vocally portray different emotional states through the production of standard utterances (e.g., numbers, letters of the alphabet, nonsense syllables, or standardized sentences). These voice samples are recorded on audiotape and later presented to lay judges, who are requested to identify or categorize the emotion expressed in each of the different portrayals.

Scherer (1989) reviewed about 30 of these studies and computed an average accuracy percentage of about 60% for studies that did not use pathological voice samples or filtered speech. This largely exceeds the percentage to be expected (approximately 12%) if listener judgments were based on guessing or other chance factors. This amount of recognition accuracy is all the more notable, in light of the fact that many of the earlier studies included emotions such as love, pride, or jealousy, which are not part of the set of basic or fundamental emotions (e.g., anger, joy, sadness, fear) generally studied in facial expression research.

More recent, methodologically well-controlled studies confirm Scherer's estimate on overall vocal emotion recognition accuracy. In a study of disgust, surprise, shame, interest, joy, fear, sadness, and anger, van Bezooijen (1984) found a mean accuracy of 65%. Based on another series of studies of five emotions (fear, joy, sadness, anger, and disgust) using different types of listener groups, Scherer, Banse, Wallbott, and Goldbeck (1991) reported a mean accuracy of 56%. Using the same stimulus material, Scherer et al. (1999) conducted a series of cross-cultural studies including eight European countries and one Asian country. They reported an accuracy percentage of 67% for the European countries and 52% for the Asian country (Indonesia). Finally, in the most comprehensive study to date, Banse and Scherer (1996) used a large stimulus set consisting of

14 emotions (in many cases, two members of one emotion family were included—e.g., hot and cold anger) spoken by 12 professional actors. When judgments of emotional voice samples from the same emotion family were aggregated (to render the data comparable to earlier work) the accuracy percentage amounted to 55%. In summary, the accurate recognition of emotion from standardized voice samples using actor portrayals seems to lie near 60%—about five times higher than what would be expected by chance. Although data from cross-cultural studies are still rare, there is some indication (see Frick, 1985; Scherer et al., 1999) that vocal expressions by members of one culture, for at least some emotions, are universally recognized by members of other cultures.

The evidence is similar to what has been found for the accurate recognition of facially expressed emotions (see Keltner & Ekman, Chapter 15, this volume), although the overall accuracy percentage for vocal emotion expressions is somewhat lower than that found in equivalent studies on the decoding of facial expressions (see Table 14.1, reproduced from Scherer, *in press*). Sadness and anger are best recognized in the voice, followed by fear. For these three emotions, the accuracy percentages are generally similar to those for facial recognition. Joy, which is very well recognized in the face, has rather mixed accuracy percentages in different studies on vocal expression, possibly due to differences with respect to whether quiet happiness or elated joy is being portrayed by the actors. Disgust is at the bottom of the scale, with the accuracy barely above chance level for the voice, although this emotion is reliably recognized in the face.

It is intriguing to speculate about the sources for these differences between emotions. As discussed in the section on the determinants of affective expression, the voice is probably more suited to the expressive and communicative

needs of certain emotions than of others. For instance, there is a clear adaptive advantage to being able to warn (fear) or threaten (anger) others in a fairly indirect way over large distances—something for which vocal expression is ideally suited. The high recognition of sadness is less easily explained, although the physiological concomitants of sadness push it in a direction away from most other emotions, so that it might be recognized through its relatively unique vocal profile. In contrast, naturally occurring vocal expressions of disgust probably consist of brief affect bursts or vocal emblems (e.g., “Yuck!”) rather than of long sentences spoken with a “disgust-specific” voice quality (see Scherer, 1994). Another possibility is that the variability in the vocal expression of disgust mirrors the diversity of the modalities involved—that is, nasal, oral, and/or visual, as well as moral evaluation (Rozin, Haidt, & McCauley, Chapter 40, this volume).

The patterns of results reported above underline the need to analyze the recognizability of different emotions separately. In addition, confusion matrices should be regularly reported, as errors are not randomly distributed and the patterns of misidentification provide important information on the judgment process. For example, the confusion matrix reported by Banse and Scherer (1996, Table 9, p. 633) provides interesting insights into the potential sources of the errors. Whereas disgust portrayals are generally confused with those of most negative emotions, there are more specific patterns for the other emotions. On the whole, as one might expect, there are many errors between members of the same emotion family. For example, hot anger is confused consistently only with cold anger and contempt; interest is confused more often with pride and happiness than with the other 11 emotions taken together. Rather than considering confusions as errors, one could interpret them as indicators of the vocal similarity

TABLE 14.1. Accuracy (%) of Facial and Vocal Emotion Recognition in Intercultural Studies

	Neutral	Anger	Fear	Joy	Sadness	Disgust	Surprise	Mean
Facial/Western/20		78	77	95	79	80	88	78
Vocal/Recent Western/11	74	77	61	57	71	31		
Facial/Non-Western/11		59	62	88	74	67	77	65
Vocal/Non-Western/1	70	64	38	28	58			

Note. Empty cells indicate that the respective emotions have not been studied in these conditions. Numbers following the slash in the first column indicate the number of cultures studied. From Scherer (1999). Copyright 1999 by Macmillan. Reprinted by permission.

ty or proximity between emotion categories. Banse and Scherer have proposed that the confusion patterns can be largely explained in terms of three dimensions of similarity: quality, intensity, and valence. However, as they have pointed out, if the three dimensions of similarity accounted for all errors, one would expect approximately symmetrical confusions between emotions, which is not always the case. For example, the substantial confusion of elation portrayals with portrayals of hot anger, panic fear, and despair is not mirrored by confusions of portrayals of those three categories with elation portrayals. It is possible that some emotions (e.g., hot anger) may be expressed by a very typical configuration of acoustic features (i.e., an emotion-specific vocal prototype), but that other emotions (e.g., elation) may lack such a unique acoustic profile. In the case of the prototypical expression of emotions, listeners probably use a top-down, pattern-matching recognition process (for an empirical analysis of top-down vs. bottom-up recognition for facial expression, see Wallbott & Ricci Bitti, 1993), whereas decoders confronted with a display of elation may have to analyze the acoustic pattern in a piecemeal, bottom-up fashion. Such bottom-up processing is more likely to be influenced by single prominent features such as high intensity, which in the case of elation makes the stimulus similar to hot anger or despair.

An alternative explanation for the asymmetry in confusions is that decoders have built-in response biases, which in the case of potentially ambiguous information lead to the selection of one emotion over another. Such biases might be based either upon the relative frequency of occurrence of different expressed emotions, or the relative adaptive importance of detecting different emotions in a speaker. It could be argued, for example, that detecting fear or anger (where one's well-being might be at stake) is more important than detecting happiness. These sort of response biases are particularly important to consider in the design of decoding studies. By giving decoders an explanation (either a true one or a cover story) of where the speech samples originated, researchers might unwittingly be installing specific response biases based upon the decoders' expectations of the most likely emotions to be encountered.

The methodology generally used in judgment studies has recently come under severe scrutiny with respect to a number of important

issues, such as the presence of explicit or implicit baselines for judgment, the use of between- and within-judgment designs (e.g., all judges judging all stimuli vs. one judge per stimulus), the role of fixed versus open responses, the type of judges used, and so forth (Russell, 1994; but see also Ekman, 1994). Although these issues need to be taken into account, there is growing evidence that under normal circumstances, judgment data are often quite robust with respect to these factors.

Identifying the Acoustic Cues Used in Emotion Inference from Voice

Given that listeners are able to recognize vocally portrayed emotions with better than chance accuracy, one should be able to determine which acoustic cues are used by listeners to attribute specific emotions to a speaker. Even when a listener's attribution is inaccurate, it is still useful to know how the misattribution might relate to the acoustic properties of the spoken expression. Different research strategies have been used to determine the importance of various acoustic cues in judging expressed emotion.

Use of Masking or Subtractive Resynthesis

The first approach consists of commencing with realistic emotion portrayals (from real life or produced by actors) and masking particular verbal/vocal cues to see how this reduces recognition of expressed emotion. Studies by Scherer et al. (1984), Scherer, Feldstein, Bond, and Rosenthal (1985), and Friend and Farrar (1994) have applied different masking techniques (filtering, randomized splicing, playing backward, pitch inversion, and tone-silence coding), each of which removes certain acoustic cues from the speech signal. The systematic use of these techniques thus allows researchers to determine which acoustic cues carry which type of emotional information. Since intelligibility is removed by all of these masking procedures, their application permits the use of natural, "real-life" speech material without biasing listener judgments with emotional verbal content. For example, the study by Scherer et al. (1984) concluded that voice quality parameters and F_0 level covaried with the strength of the judged emotion, conveying emotion independently of the verbal context.

However, different intonation contour types were found to interact with sentence grammar in an all-or-nothing, configurational manner, and conveyed more attitudinal information.

Use of Additive Resynthesis or Synthesis

Recently the quality of speech synthesis has become good enough to be used to systematically manipulate different acoustic cues, starting from a neutral synthesized utterance, in an attempt to produce synthetic emotional expressions. In an early study, Lieberman and Michaels (1962) studied the effect of variations of F_0 and envelope contour on emotion inference. Scherer and Oshinsky (1977) studied the effects of several cues (amplitude variation; pitch level, contour, and variation; tempo; envelope; harmonic richness; tonality; and rhythm) on emotion attributions to sentence-like sound sequences and musical melodies. More recent work with the aim of adding emotional expressivity to text-to-speech systems (Murray & Arnott, 1993) has met with mixed success, but nevertheless shows great promise if it can be done in a systematic manner.

Resynthesis techniques make it possible to take neutral natural voices and systematically change different cues via digital manipulation of the sound waves. In a number of studies by Scherer and his collaborators, resynthesis of real utterances has been used to systematically manipulate F_0 level, contour, variability, and range, as well as intensity, duration, and accent structure (Ladd, Silverman, Tolkmitt, Bergmann, & Scherer, 1985; Tolkmitt, Bergmann, Goldbeck, & Scherer, 1988). Listener judgments of apparent speaker attitude and emotional state for each stimulus showed strong direct effects for all of the variables manipulated, with relatively few effects due to interactions between the manipulated variables. Of all variables studied, F_0 range affected the judgments the most, with narrow F_0 range being judged as signaling sadness and wide F_0 range being judged as expressing high-arousal, negative emotions such as annoyance or anger. High speech intensity was also interpreted as a sign of negative emotions or aggression. Fast speech led to inferences of joy, with slow speech judged as a mark of sadness. Only very minor effects for speaker and utterance content were found, indicating that the results are likely to generalize over different speakers and utterances.

Use of Unaltered Emotion Portrayals

It is possible to use acoustic analyses to measure the acoustic characteristics of vocal emotion portrayals (acted or natural), and then to correlate these with the listeners' judgments of the underlying emotion or attitude of the speaker. Several studies of this type have yielded information on which vocal characteristics affect the judges' inference (Scherer, Rosenthal, & Koivumaki, 1972; van Bezooijen, 1984). In one of the most recent and most thorough studies of this type, Banse and Scherer (1996) showed that a large proportion of the variance in judgments of emotion was explained by a set of about 9–10 acoustic measures, including mean F_0 , standard deviation of F_0 , mean energy, duration of voiced periods, proportion of energy up to 1,000 Hz, and spectral dropoff. Table 14.2 (adapted from Banse & Scherer, 1996, Table 8, p. 632) shows the direction (i.e., positive or negative) of beta coefficients resulting from regressing judges' emotion ratings on the acoustical parameters, as well as the total multiple correlation coefficients. The data indicate that quite specific patterns of acoustic parameters are used to infer different emotional states.

Comparison of Human Judges with Statistical Discrimination

Banse and Scherer (1996, Table 9, p. 633) also compared the performance of human judges to statistical algorithms such as discriminant analysis and jackknifing. Such an approach provides a test of whether the acoustic measures correlated to listener judgments do indeed discriminate the different expressed emotions. If the set of acoustic cues being analyzed is relatively similar to the set being used by listeners, one would expect statistical routines to approach human levels of discrimination. Banse and Scherer found not only that the discrimination rates were very similar, but that there was a high correspondence between the confusion matrices. The use of more sophisticated classification routines, particularly those that allow for easy interpretation of decision rules (such as decision tree algorithms), should shed more light on the way human judges infer speaker state from acoustic characteristics of speech.

Regardless of the analytic and statistical methods used in decoding studies, there is also a need to progress from simply measuring listeners' ability to discriminate between different

TABLE 14.2. Direction of Beta Weights, and Multiple Correlations, Resulting from Regressing Judges' Emotion Ratings on Acoustical Parameters

	Mean F_0	SD F_0	Mean energy	Voiced duration	HI	LFE	R
Hot anger	+	+			—		.63***
Cold anger							.16
Panic	+						.39***
Anxiety	+	—	—				.38***
Despair	+	—		+	+		.44***
Sadness			—			+	.49***
Elation	+					+	.39***
Happiness							.27*
Interest							.18
Boredom	—		+	+			.40***
Shame			—				.38***
Pride	—						.28*
Disgust						—	.17
Contempt	—	+					.36***

Note. Empty cells indicate beta weights not significantly different from zero. HI, Hammarberg index; LFE, proportion of voiced energy up to 1,000 Hz. Adapted from Banse and Scherer (1996). Copyright 1996 by the American Psychological Association. Adapted by permission.

* $p \leq .05$. *** $p \leq .001$.

expressed emotions, toward measuring the perceived quality, reality or authenticity of emotional expressions. Although a listener may be quite able to identify the intended emotion expressed in a spoken sample, this could reflect simply the use of stereotypical emotion prototypes by the speaker, rather than realistic emotion portrayal. This is obviously an important consideration when acted emotions are used in research, but it is equally important in other settings, when for a variety of reasons (e.g., task demands, social pressures) speakers may be acting to some extent. Even if one regards such acted emotional expressions as valid in their own right, it still seems important to examine how these expressions are perceived, compared with more natural affective speech.

CONCLUSIONS

This chapter has highlighted some of the many evolutionary, physiological, cognitive, social, and cultural factors that shape the way humans express and perceive emotions in speech. With such a large and seemingly disparate number of determinants, it might seem as if the topic were too messy to allow us to expect any invariance in empirical findings. Perhaps surprisingly, however, our summary of research into the production and perception of emotional speech has

revealed considerable consistency. On the production side, the evidence is starting to accumulate that humans consistently modify their speech in specific ways to express different emotions. Results of perception studies indicate that emotions expressed in speech are to a large extent successfully detected by a variety of populations, on the basis of an experimentally identifiable set of acoustic parameters. One reason for the consistency in the results is no doubt that most research to date has been limited to settings in which many of the factors described above have been eliminated or controlled for. In addition to further refinement of analysis techniques, and a focus on real as well as acted emotional speech, there is clearly a need for studies that better quantify the relative contribution of culture, language, and social strategy to the vocal communication of emotion. To address these issues in a manner that allows results from different studies to be integrated and compared, a coordinated, interdisciplinary approach to research on the vocal communication of emotion will be required.

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